Weierstrass Gamma Function by Primes

Vladislav Panin

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Abstract

We introduce a Weierstrass-type gamma function defined by a product over the primes, derive an integral representation involving the prime-counting function $\pi(x)$, and formulate a conjecture on the location of zeros in the strip $0 < \mathrm{Re}(z) < 1$. Numerical approximations of the first zeros on the line $\mathrm{Re}(z) = \frac{1}{2}$ are presented.

Definition and Integral Representation

The Weierstrass gamma function by primes:

$$\frac{1}{G_p(z)} = z \prod_{p}^{\infty} \left(1 + \frac{z}{p}\right) e^{-z/p}$$

$$\frac{1}{G_p(z)} = z \exp\left(-\int_2^{\infty} \pi(x) \frac{z^2}{x^2(x+z)}\, dx\right)$$

Remark: The integral converges absolutely for $z \in \mathbb{C} \setminus (-\infty, -2]$.

$$\Phi(z) = \frac{G_p\left(\frac{1-z}{2}\right)}{G_p\left(\frac{z}{2}\right)}$$

Conjecture: All zeros of $f(z) = 1 - \Phi(z)$ with $0 < \mathrm{Re}(z) < 1$ lie on the line $\mathrm{Re}(z) = \frac{1}{2}$. The zeros are located at $z = \frac{1}{2} + it$, $t \in \mathbb{R}$:

$$t \in \{7.5593242,\ 14.6100647,\ 20.3194037,\ 25.5268164,\ 30.4450039,\ 35.1672954,\ 39.7447722,\ 44.2091084,\ 48.5815992,\ 52.8774028,\ 57.1077783,\ 61.2813706,\ 65.4049958,\ 69.4841471,\ 73.5233328,\ 77.5263103,\ \dots\}$$

References

[1] Fröberg, C. E. (1968). On the prime zeta function. *BIT Numerical Mathematics*, 8(3), 187–202.

[2] Apostol, T. M. (1976). *Introduction to Analytic Number Theory*. Springer-Verlag.